

KTH-TIPS2 SPD Geometry Analysis

Summary of Results

This report summarizes the analysis of SPD (Symmetric Positive Definite) manifold geometries applied to image covariance descriptors from the KTH-TIPS2-b texture dataset. Three Riemannian metrics --Affine Invariant (AI), Log-Euclidean (LogE), and Bures-Wasserstein (BW) --are compared across visualization, feature preprocessing, and downstream classification tasks.

Dataset: KTH-TIPS2-b -- 11 material texture classes, 22x22 covariance matrices (253-dimensional tangent features after projection). Subset used: **sample_a** ($n \sim 1,188$).

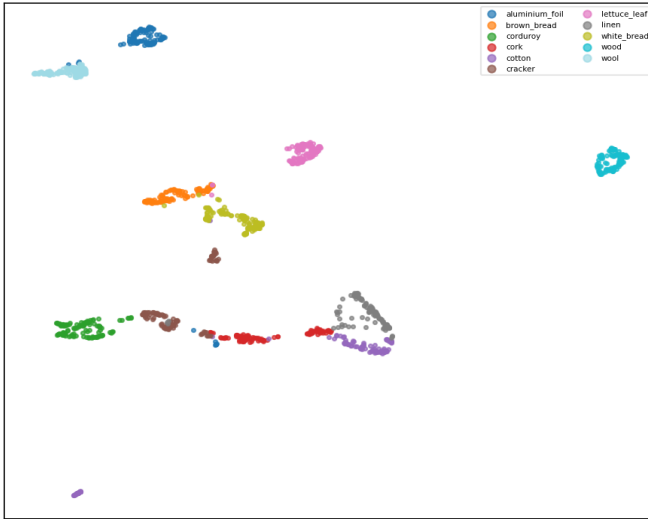
Pipeline overview: Raw images -> covariance matrices -> SPD geometry selection -> barycenter + tangent projection -> optional standardization -> optional dimension reduction (PCA / UMAP) -> classification.

Task 1: Manifold UMAP vs Tangent-Space UMAP

Two UMAP embedding strategies are compared for each geometry: **Manifold UMAP** operates on precomputed pairwise SPD distances, while **Tangent UMAP** first projects matrices into tangent space and then applies standard Euclidean UMAP. Neighborhood overlap ($k=15$), trustworthiness, and Spearman rank correlation between distance matrices quantify agreement.

Task 1: Manifold UMAP vs Tangent-Space UMAP

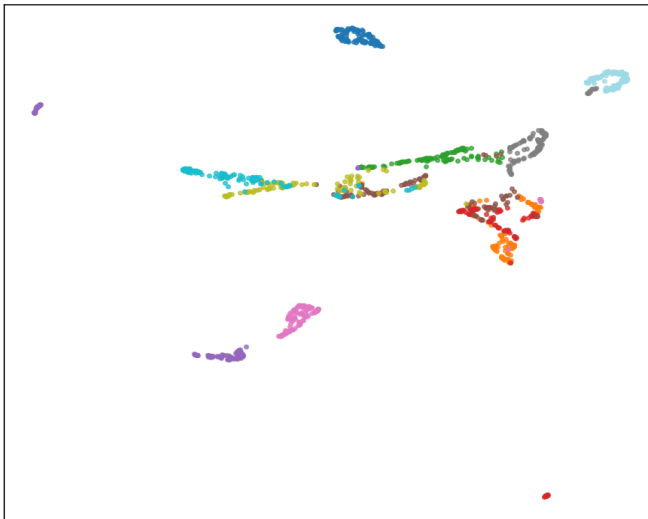
BW - Manifold UMAP



BW - Tangent UMAP



AI - Manifold UMAP



AI - Tangent UMAP



LOGE - Manifold UMAP



LOGE - Tangent UMAP



Figure 1: UMAP embeddings colored by material class. Left column: Manifold UMAP; Right column: Tangent UMAP. Rows: BW, AI, LogE (top to bottom).

Quantitative Comparison Metrics

Metric	Neighborhood Overlap	Trustworthiness (Manifold)	Trustworthiness (Tangent)	Spearman rho	p-value
BW	0.8539	0.9796	0.9795	0.9429	0.0
AI	0.4781	0.9929	0.9832	0.7118	0.0
LOGE	0.9386	0.9885	0.9891	0.9956	0.0

Key findings:

- **LogE** shows the highest agreement between manifold and tangent embeddings (overlap = 0.939, Spearman rho = 0.996), indicating its tangent projection preserves manifold structure almost perfectly.
- **BW** shows strong agreement (overlap = 0.854, rho = 0.943), with both embeddings achieving comparable trustworthiness (~0.980).
- **AI** has markedly lower neighborhood overlap (0.478) and Spearman correlation (0.712), suggesting that tangent projection under the AI metric distorts local neighborhood structure more substantially than the other geometries.
- All six embeddings achieve high trustworthiness (≥ 0.979), confirming UMAP faithfully preserves local neighborhoods regardless of input representation.

Task 2: Raw vs Standardized Tangent Features

This task investigates the effect of z-score standardization on tangent-space features across three questions: (Q1) how feature variance distributions change, (Q2) how PCA explained variance is affected, and (Q3) whether standardization improves classification accuracy.

Q1: Feature Standard Deviations

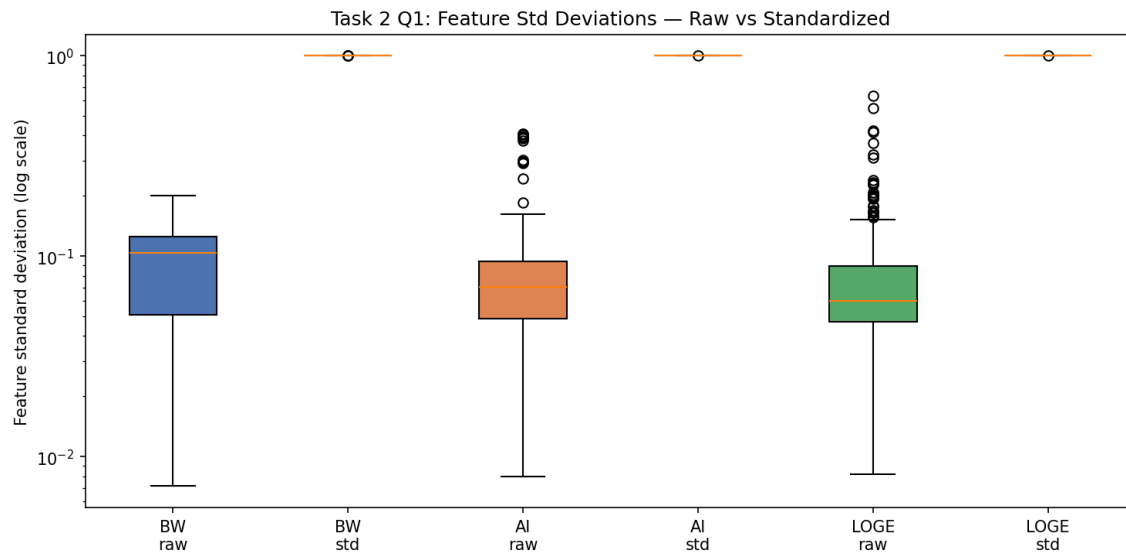


Figure 2: Distribution of per-feature standard deviations before and after standardization (log scale).

Raw feature variances span several orders of magnitude, especially for AI and BW. Standardization collapses all features to unit variance, equalizing their influence on distance-based methods.

Q2: PCA Cumulative Variance Explained

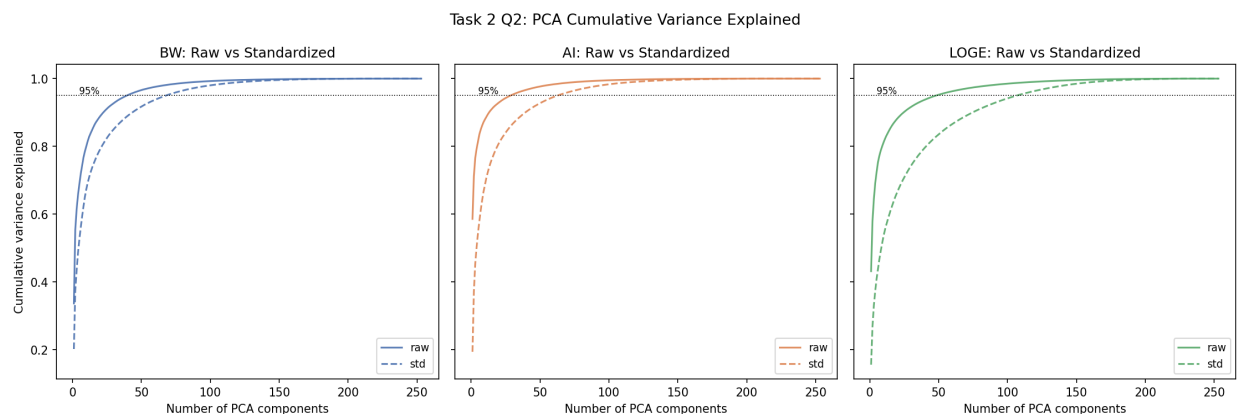


Figure 3: Cumulative variance explained by PCA components. Solid = raw, dashed = standardized.

For raw features, a small number of PCA components capture most of the variance (dominated by high-variance features). After standardization, variance is spread more evenly, requiring more components to reach 95%. This means PCA on raw features is driven by scale rather than discriminative power.

Q3: Classification Accuracy (KNN, $k=5$)

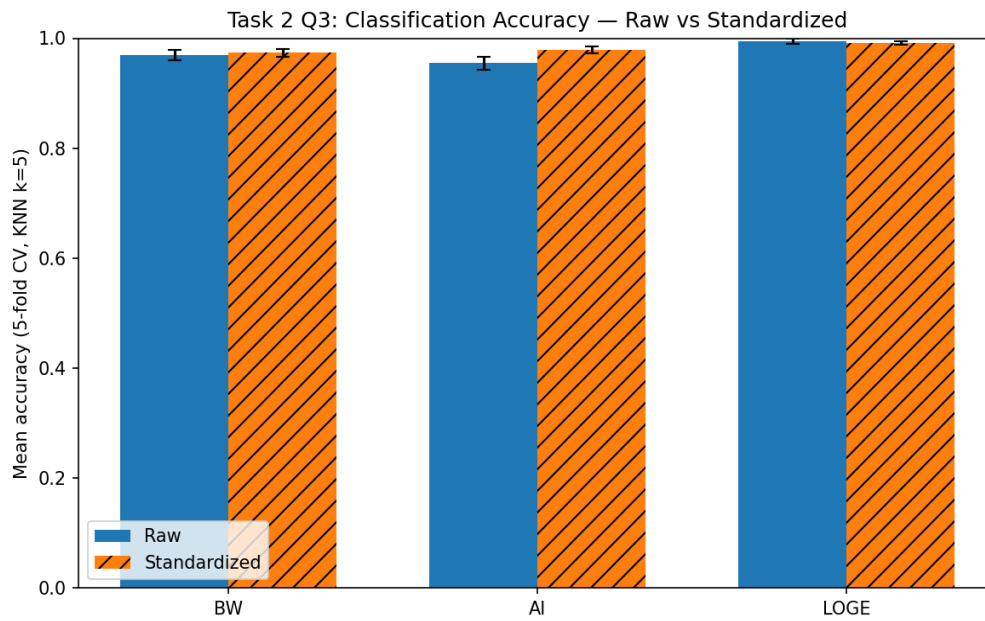


Figure 4: Mean 5-fold CV accuracy with error bars (KNN k=5), raw vs standardized.

Metric	Standardized	Accuracy (mean)	Accuracy (std)
BW	No	0.9697	0.0094
BW	Yes	0.9739	0.0072
AI	No	0.9545	0.0115
AI	Yes	0.9789	0.006
LOGE	No	0.995	0.0049
LOGE	Yes	0.9907	0.0032

- Standardization improves AI accuracy substantially (95.5% -> 97.9%), consistent with AI's wide raw variance spread.
- BW sees a modest gain (97.0% -> 97.4%).
- LogE is slightly hurt by standardization (99.5% -> 99.1%), suggesting its raw feature scales are already well-balanced for KNN.

Task 3: Downstream Classification -- Nested Cross-Validation

A rigorous nested CV design (5 outer / 3 inner folds) evaluates 54 combinations: 3 geometries x 3 pipelines x 2 standardization settings x 3 classifiers. Pipeline A uses full 253-dim tangent features, Pipeline B applies PCA, and Pipeline C applies UMAP for dimension reduction. Latent dimensions q in $\{5, 10, 20\}$ are tuned in the inner loop along with classifier hyperparameters.

Top 10 Configurations by Accuracy

Metric	Pipeline	Std	Classifier	Accuracy	$\pm$ Std	AUC
LOGE	A	Yes	logreg	0.9992	0.0017	1.0000
LOGE	A	Yes	svc	0.9992	0.0017	1.0000
LOGE	A	No	logreg	0.9983	0.0021	1.0000
LOGE	A	No	svc	0.9983	0.0034	1.0000
LOGE	B	Yes	svc	0.9983	0.0021	1.0000
AI	A	No	svc	0.9975	0.0034	0.9999
LOGE	B	No	svc	0.9975	0.0021	1.0000
AI	A	Yes	svc	0.9966	0.0049	1.0000
BW	A	No	svc	0.9966	0.0031	1.0000
BW	A	Yes	svc	0.9966	0.0017	1.0000

Key findings:

- **LogE dominates:** the top configurations are all LogE with Pipeline A (full features), achieving 99.8–99.9% accuracy and perfect AUC (1.000). LogE + standardized logistic regression and SVM both reach 99.9%.
- **Pipeline A (no dimension reduction) consistently outperforms B and C** across all geometries. The 253-dimensional tangent features contain sufficient discriminative information without needing reduction.
- **Pipeline C (UMAP reduction) consistently underperforms**, likely because UMAP's nonlinear mapping discards information useful for supervised classification, despite being good for visualization.
- **SVM and logistic regression outperform KNN** across all settings, with SVM generally achieving the highest accuracy per geometry-pipeline combination.
- **Runtime:** LogE is dramatically faster than BW and AI (e.g., 4s vs 650s per Pipeline A combination), making it the most practical choice in addition to being the most accurate.

Conclusions

1. **Log-Euclidean geometry is the clear winner** for this dataset, achieving the highest classification accuracy (99.9%), the best manifold–tangent agreement (Task 1), and the fastest computation time. Its tangent projection preserves virtually all discriminative structure from the manifold.

2. **Tangent projection is a reliable proxy for manifold distances**, especially under LogE and BW geometries. The AI metric shows more distortion in tangent space, which may explain its slightly lower classification performance.

3. **Standardization helps when raw feature scales are imbalanced** (AI benefits most), but can slightly hurt when scales are already balanced (LogE). The choice should be geometry-dependent.

4. **Dimension reduction is unnecessary for this feature set**. Full 253-dim tangent features with a simple SVM or logistic regression achieve near-perfect classification. PCA and UMAP reduction reduce accuracy, with UMAP showing the largest drop.